

Linear models are (almost) all you need to know

Advanced Statistics and Data Analysis

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① ANOVA

② The t-test and other tests

③ The QR decomposition

④ Interactions

# ANOVA

## Analysis of Variance (ANOVA)

Last semester you have seen ANOVA as a generalization of the t-test for more than two groups.

Let's recall it with an example.

```
select(mtcars, mpg, cyl) |>  
  head()
```

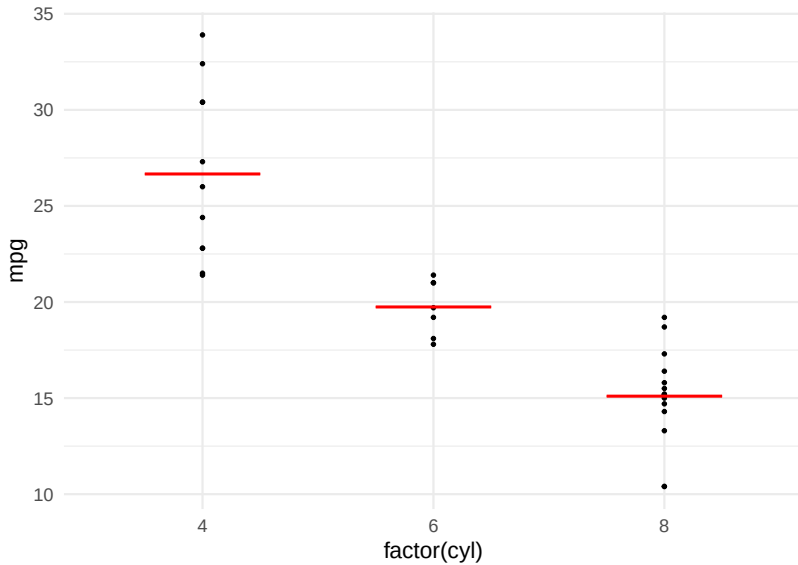
|                   | mpg  | cyl |
|-------------------|------|-----|
| Mazda RX4         | 21.0 | 6   |
| Mazda RX4 Wag     | 21.0 | 6   |
| Datsun 710        | 22.8 | 4   |
| Hornet 4 Drive    | 21.4 | 6   |
| Hornet Sportabout | 18.7 | 8   |
| Valiant           | 18.1 | 6   |

Does car efficiency, measured in miles-per-gallon (mpg), depend on the type of engine?

In our data, we have three engine types: 4, 6, and 8 cylinder engines.

We can use a one-way ANOVA to test whether the efficiency is the same or different across cars with these engines.

## ANOVA example



## ANOVA example

```
anova <- aov(mpg ~ factor(cyl), data=mtcars)
summary(anova)
```

|             | Df | Sum Sq | Mean Sq | F value | Pr(>F)       |
|-------------|----|--------|---------|---------|--------------|
| factor(cyl) | 2  | 824.8  | 412.4   | 39.7    | 4.98e-09 *** |
| Residuals   | 29 | 301.3  | 10.4    |         |              |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

# ANOVA as a linear model

In fact, we can perform the same analysis as a linear regression, choosing `mpg` as the response and `cyl` as a *categorical* covariate.

Remember that for categorical covariates, we need to define *dummy variables*.

In the case of `cyl`, which has three levels, we need two dummies (plus the intercept).



## Warning

- Why do we need only two dummy variables?
- What would happen if we include all three in the regression?



# ANOVA as a linear model

```
fit <- lm(mpg ~ factor(cyl), data=mtcars)
summary(fit)
```

Call:

```
lm(formula = mpg ~ factor(cyl), data = mtcars)
```

Residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -5.2636 | -1.8357 | 0.0286 | 1.3893 | 7.2364 |

Coefficients:

|              | Estimate | Std. Error | t value | Pr(> t )     |
|--------------|----------|------------|---------|--------------|
| (Intercept)  | 26.6636  | 0.9718     | 27.437  | < 2e-16 ***  |
| factor(cyl)6 | -6.9208  | 1.5583     | -4.441  | 0.000119 *** |
| factor(cyl)8 | -11.5636 | 1.2986     | -8.905  | 8.57e-10 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.223 on 29 degrees of freedom

Multiple R-squared: 0.7325, Adjusted R-squared: 0.714

F-statistic: 39.7 on 2 and 29 DF, p-value: 4.979e-09

## ANOVA as a linear model

As we can see from the F statistic and its p-value, the results are *exactly the same*.  
This is not by coincidence, but because we are performing *exactly the same analysis*.  
In fact, R's `aov` function internally calls `lm.fit` exactly like `lm`.

## The design matrix

To understand mathematically why this is the same thing, we have to introduce the *design matrix*, which is just another name for the matrix  $X$  whose columns contain the covariates that we include in the model.

When we perform a regression against categorical variables, `lm` implicitly defines a set of dummy variables to include in the design matrix.

We can see what happens internally, by using the `model.matrix` function, which transforms the formula we feed into `lm` into a design matrix.

## The design matrix

```
head(model.matrix(~factor(cyl), data=mtcars))
```

|                   | (Intercept) | factor(cyl)6 | factor(cyl)8 |
|-------------------|-------------|--------------|--------------|
| Mazda RX4         | 1           | 1            | 0            |
| Mazda RX4 Wag     | 1           | 1            | 0            |
| Datsun 710        | 1           | 0            | 0            |
| Hornet 4 Drive    | 1           | 1            | 0            |
| Hornet Sportabout | 1           | 0            | 1            |
| Valiant           | 1           | 1            | 0            |

## Why not all three dummies?

Remember that we need to invert the design matrix  $X$  to obtain the least squares solution:

$$\hat{\beta} = (X^{\top} X)^{-1} X^{\top} y.$$

This means that  $X^{\top} X$  needs to be invertible, which implies that  $X$  is *full rank*.

### ! Definition

The rank of a matrix is the number of columns that are independent of all the others.

If the rank is equal to the number of columns, the matrix is said to be full rank.

## Why not all three dummies?

```
X <- cbind(1, model.matrix(~factor(cyl)-1, data=mtcars))  
head(X)
```

|                   |   | factor(cyl)4 | factor(cyl)6 | factor(cyl)8 |
|-------------------|---|--------------|--------------|--------------|
| Mazda RX4         | 1 | 0            | 1            | 0            |
| Mazda RX4 Wag     | 1 | 0            | 1            | 0            |
| Datsun 710        | 1 | 1            | 0            | 0            |
| Hornet 4 Drive    | 1 | 0            | 1            | 0            |
| Hornet Sportabout | 1 | 0            | 0            | 1            |
| Valiant           | 1 | 0            | 1            | 0            |

```
qr(X)$rank
```

```
[1] 3
```

```
solve(crossprod(X))
```

```
Error in solve.default(crossprod(X)): system is computationally singular: reciprocal condition number
```

## ANOVA and linear regression: two equivalent views

When we think about ANOVA, we are thinking about comparing the means of three or more groups.

In the `mtcars` example, we have three groups, and we can think about the following model

$$y_i = \begin{cases} \mu_1 + \varepsilon_i & \text{if obs. } i \text{ is in group 1} \\ \mu_2 + \varepsilon_i & \text{if obs. } i \text{ is in group 2} \\ \mu_3 + \varepsilon_i & \text{if obs. } i \text{ is in group 3} \end{cases}$$

# ANOVA and linear regression: two equivalent views

In matrix notation:

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_i \\ y_{i+1} \\ \vdots \\ y_{n-1} \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ \vdots & & \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ \vdots & & \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_i \\ \varepsilon_{i+1} \\ \vdots \\ \varepsilon_{n-1} \\ \varepsilon_n \end{bmatrix}$$



# ANOVA and linear regression: two equivalent views

In linear regression, we define the model as

$$y_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} + \varepsilon_i,$$

where  $p$  is the number of covariates in  $X$ .

# ANOVA and linear regression: two equivalent views

In matrix notation:

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_i \\ y_{i+1} \\ \vdots \\ y_{n-1} \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ \vdots & & \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ \vdots & & \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_i \\ \varepsilon_{i+1} \\ \vdots \\ \varepsilon_{n-1} \\ \varepsilon_n \end{bmatrix}$$

# ANOVA and linear regression: two equivalent views

Mathematically, we can see that

$$y = \beta_0 X^{(1)} + \beta_1 X^{(2)} + \beta_2 X^{(3)} + \varepsilon,$$

where  $X^{(j)}$  indicates the  $j$ -th column of the design matrix.

Hence,

$$y_i = \begin{cases} \beta_0 & \text{if obs. } i \text{ is in group 1} \\ \beta_0 + \beta_1 & \text{if obs. } i \text{ is in group 2} \\ \beta_0 + \beta_2 & \text{if obs. } i \text{ is in group 3} \end{cases}$$

# ANOVA and linear regression: two equivalent views

It is now easy to see that:

- $\mu_1 = \beta_0$ , i.e., the mean of the *reference group*;
- $\mu_2 = \beta_0 + \beta_1 \implies \beta_1 = \mu_2 - \mu_1$ ;
- $\mu_3 = \beta_0 + \beta_2 \implies \beta_2 = \mu_3 - \mu_1$ ;

This gives an intuitive interpretation of the regression coefficients in the case of dummy variables.

## ANOVA and linear regression: two equivalent views

We can check this in R.

```
mtcars |> group_by(cyl) |> summarize(mean(mpg))
```

```
# A tibble: 3 x 2
  cyl `mean(mpg)`
  <dbl>         <dbl>
1     4         26.7
2     6         19.7
3     8         15.1
```

```
unnamed(c(fit$coefficients[1],
fit$coefficients[1] + fit$coefficients[2],
fit$coefficients[1] + fit$coefficients[3]))
```

```
[1] 26.66364 19.74286 15.10000
```

But what is the average difference between the MPG of 6 and 8 cylinder cars?

Is this difference statistically significant?

To answer this question, we need to compute a linear combination of the parameters, called *a contrast*.

In fact,

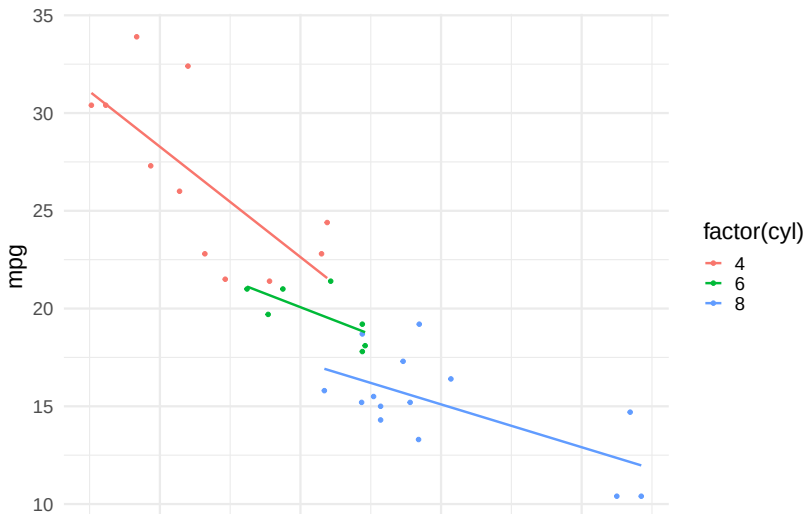
$$\mu_3 - \mu_2 = (\beta_0 + \beta_2) - (\beta_0 + \beta_1) = \beta_2 - \beta_1$$

We can apply the usual t-test to test the significance of *any linear combination of the coefficients*.

**Add example in R**

## Advantages of linear modeling

The advantage of the linear model formulation is that we can now include multiple covariates, including continuous ones, and even interactions.



## Including weight in the model

If we include weight in the model, we have to include an additional column in the design matrix.

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_i \\ y_{i+1} \\ \vdots \\ y_{n-1} \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & w_1 \\ 1 & 0 & 0 & w_2 \\ \vdots & & & \\ 1 & 1 & 0 & w_i \\ 1 & 1 & 0 & w_{i+1} \\ \vdots & & & \\ 1 & 0 & 1 & w_{n-1} \\ 1 & 0 & 1 & w_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_i \\ \varepsilon_{i+1} \\ \vdots \\ \varepsilon_{n-1} \\ \varepsilon_n \end{bmatrix}$$



## Including weight in the model

Now we have:

$$y_i = \begin{cases} \beta_0 + \beta_3 w_i & \text{if obs. } i \text{ is in group 1} \\ \beta_0 + \beta_1 + \beta_3 w_i & \text{if obs. } i \text{ is in group 2} \\ \beta_0 + \beta_2 + \beta_3 w_i & \text{if obs. } i \text{ is in group 3} \end{cases}$$

This means that the relationship between MPG and weight has the same slope across groups, but start from a different intercept.

## Including weight in the model

Call:

```
lm(formula = mpg ~ factor(cyl) + wt, data = mtcars)
```

Residuals:

|  | Min     | 1Q      | Median  | 3Q     | Max    |
|--|---------|---------|---------|--------|--------|
|  | -4.5890 | -1.2357 | -0.5159 | 1.3845 | 5.7915 |

Coefficients:

|              | Estimate | Std. Error | t value | Pr(> t )     |
|--------------|----------|------------|---------|--------------|
| (Intercept)  | 33.9908  | 1.8878     | 18.006  | < 2e-16 ***  |
| factor(cyl)6 | -4.2556  | 1.3861     | -3.070  | 0.004718 **  |
| factor(cyl)8 | -6.0709  | 1.6523     | -3.674  | 0.000999 *** |
| wt           | -3.2056  | 0.7539     | -4.252  | 0.000213 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.557 on 28 degrees of freedom

Multiple R-squared: 0.8374, Adjusted R-squared: 0.82

F-statistic: 48.08 on 3 and 28 DF, p-value: 3.594e-11

## Including interactions

If we include the interactions in the model, we have to include two additional columns in the design matrix, which are obtained by multiplying the two relative columns.

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_i \\ y_{i+1} \\ \vdots \\ y_{n-1} \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & w_1 & 0 & 0 \\ 1 & 0 & 0 & w_2 & 0 & 0 \\ \vdots & & & & & \\ 1 & 1 & 0 & w_i & w_i & 0 \\ 1 & 1 & 0 & w_{i+1} & w_{i+1} & 0 \\ \vdots & & & & & \\ 1 & 0 & 1 & w_{n-1} & 0 & w_{n-1} \\ 1 & 0 & 1 & w_n & 0 & w_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_i \\ \varepsilon_{i+1} \\ \vdots \\ \varepsilon_{n-1} \\ \varepsilon_n \end{bmatrix}$$

## Including interactions

Call:

```
lm(formula = mpg ~ factor(cyl) * wt, data = mtcars)
```

Residuals:

|  | Min     | 1Q      | Median  | 3Q     | Max    |
|--|---------|---------|---------|--------|--------|
|  | -4.1513 | -1.3798 | -0.6389 | 1.4938 | 5.2523 |

Coefficients:

|                 | Estimate | Std. Error | t value | Pr(> t ) |     |
|-----------------|----------|------------|---------|----------|-----|
| (Intercept)     | 39.571   | 3.194      | 12.389  | 2.06e-12 | *** |
| factor(cyl)6    | -11.162  | 9.355      | -1.193  | 0.243584 |     |
| factor(cyl)8    | -15.703  | 4.839      | -3.245  | 0.003223 | **  |
| wt              | -5.647   | 1.359      | -4.154  | 0.000313 | *** |
| factor(cyl)6:wt | 2.867    | 3.117      | 0.920   | 0.366199 |     |
| factor(cyl)8:wt | 3.455    | 1.627      | 2.123   | 0.043440 | *   |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.449 on 26 degrees of freedom

Multiple R-squared: 0.8616, Adjusted R-squared: 0.8349

F-statistic: 32.36 on 5 and 26 DF, p-value: 2.258e-10

Now we have:

$$y_i = \begin{cases} \beta_0 + \beta_3 w_i & \text{if obs. } i \text{ is in group 1} \\ (\beta_0 + \beta_1) + (\beta_3 + \beta_4) w_i & \text{if obs. } i \text{ is in group 2} \\ (\beta_0 + \beta_2) + (\beta_3 + \beta_5) w_i & \text{if obs. } i \text{ is in group 3} \end{cases}$$

This means that the relationship between MPG and weight changes both in terms of slope and intercept across groups.

**!** Important

How do you interpret the parameters in this case?

## The t-test and other tests

Consider introducing ranks and show that the nonparametric tests can also be written as linear models (perhaps as exercise/homework).

## The QR decomposition



# Interactions